ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** bhukampa	Bhukampa means shaking of the Earth
2	**(a)** a natural feature of the Earth's surface	e.g. mountain, plateau
3	**(a)** 2001	Major Indian earthquake
4	**(d)** thinner, about 5-7 km	Oceanic crust
5	**(d)** the top of a sea arch collapses leaving a pillar	Coastal erosion
6	**(a)** transform boundary	Plates slide past
7	**(c)** fjord	Sea floods glacial valley
8	**(a)** Nagaland	Earthen bunds on hillslopes
9	**(a)** lithosphere	Broken into plates
10	**(c)** convection currents in the mantle	Heat from core

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Theory that crust is broken into plates (1); Plates move slowly over the semi-molten layer (1).	1+1
12	Natural feature of the Earth's surface (1); Mountains and plateaus (1).	1+1
13	Heat from the core (1); Creates convection currents in the mantle (1).	1+1
14	Weathering breaks rocks in place (1); Erosion carries material away (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Crust is thin outer layer (1); Mantle is thick and hot (1); Core is iron-nickel, outer liquid and inner solid (1).	1+1+1
16	Continental plates carry continents (1); Oceanic plates carry ocean floors (1); Mixed plates carry both (1).	1+1+1
17	Running water: valley (1); Glacier: U-shaped valley, wind: dune (1); Waves: cliffs, groundwater: caves (1).	1+1+1
	OR: Stalactite hangs from the cave ceiling (1); Stalagmite rises from the cave floor (1); Formed in limestone caves (1).	1+1+1
18	Deforestation (1); Mining and road building (1); Unplanned construction on slopes (1).	1+1+1
	OR: River slows at its mouth (1); Deposits sediment in a fan shape (1); Distributaries spread out (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Crust: 30-40 km under continents, 5-7 km under oceans (1); Lithosphere: crust plus upper mantle, broken into plates (1); Asthenosphere: partly molten layer (1); Mantle: thick, hot and mostly solid (1); Core: liquid outer core, solid inner core (1).	1+1+1+1+1
	OR: Farmers lose topsoil (1); Coasts and rivers lose land (1); Contouring and bunding (1); Terracing and check dams (1); Zabo system of Nagaland (1).	1+1+1+1+1
20	Yardang and ventifact (1); Deflation hollow and desert pavement (1); Dunes of four types (1); Dunes check desertification (1); Supply sand and adventure sports (1).	1+1+1+1+1
	OR: Crust broken into plates (1); Movement caused by convection currents (1); Convergent forms fold mountains (1); Divergent forms ridges (1); Transform causes earthquakes (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).
- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.

- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).
- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.
- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Erosion	Erosion carries material away
2	**(b)** convergent boundary	Continental collision
3	**(a)** weathering	In-place breakdown
4	**(c)** movement of the Earth's crust	Moving plates
5	**(b)** barchan	Single wind direction
6	**(a)** asthenosphere	Allows plates to move
7	**(a)** terraces	Terracing
8	**(a)** 30-40 km thick	Continental crust is thicker than oceanic crust
9	**(a)** cirque	Glacial landform
10	**(c)** medial	Joins lateral moraines

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Depression where ground collapses into a cavity (1); Found in limestone areas (1).	1+1
12	A winding bend of a river (1); Erosion on outer bank and deposition on inner bank (1).	1+1
13	Debris deposited by a glacier (1); Lateral, medial or terminal (1).	1+1
14	Physical and chemical (1); Biological (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fertile soil for rice and jute (1); Fishing and trade (1); Dense settlement, but flood-prone (1).	1+1+1
16	Waves erode a headland to make caves (1); Caves join to form an arch (1); Arch collapses leaving a stack (1).	1+1+1
17	Fertile topsoil is removed (1); Land, houses and roads are washed away (1); Yields and livelihoods fall (1).	1+1+1
	OR: Convergent - Himalaya (1); Divergent - Mid-Atlantic Ridge (1); Transform - San Andreas Fault (1).	1+1+1
18	Changes in wind and rain (1); Cloud and animal behaviour (1); Planetary alignments in the Brihatsamhita (1).	1+1+1
	OR: Plates collide, separate or slide there (1); Stress and heat release (1); Seen in the Ring of Fire (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rivers supported farming societies (1); Himalayas shielded India yet allowed passes (1); Thar and Silk Route trade (1); Coasts supported trade (1); Khyber Pass exchange (1).	1+1+1+1+1
	OR: Upper: V valleys, waterfalls (1); Middle: meanders, oxbow lakes (1); Lower: deltas, levees (1); Erosion and deposition vary (1); Importance for settlements (1).	1+1+1+1+1
20	Weathering breaks rocks at surface (1); Physical: temperature and frost (1); Chemical: reaction with water and air (1); Biological: plants and animals (1); Weathering forms soil (1).	1+1+1+1+1
	OR: Heavy rain and slopes (1); Deforestation and mining (1); Snowfall and warming trigger avalanches (1); Afforestation and drainage (1); Early warning (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).
- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.
- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).

- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.
- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** stalactites	Karst
2	**(c)** inner core	Solid metal ball
3	**(a)** liquid iron and nickel	Fluid layer
4	**(c)** crust	Surface layer
5	**(c)** delta	Sediment deposit
6	**(a)** convergent	Collision or subduction
7	**(c)** upper course of a river	Steep gradient
8	**(c)** contour ploughing	Contour ploughing
9	**(c)** middle and lower course	The river has less energy and swings sideways
10	**(c)** delta	Large delta

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Crust and mantle (1); Outer core and inner core (1).	1+1
12	Zone around the Pacific Ocean (1); Most earthquakes and volcanoes occur here (1).	1+1
13	Earthen embankments along contour lines (1); Slow run-off and reduce erosion (1).	1+1
14	Convergent: plates move towards each other (1); Divergent: plates move apart (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fertile soil (1); Natural dams and lakes (1); Water for irrigation and power (1).	1+1+1
16	Contouring: trenches on contour lines (1); Bunding: earthen embankments (1); Terracing: level steps on hillside (1).	1+1+1
17	Steep gradient with strong erosion (1); V-shaped valleys (1); Waterfalls and rapids (1).	1+1+1
	OR: U-shaped valley (1); Cirque (1); Arete (1).	1+1+1
18	Physical: temperature change and frost (1); Chemical: minerals react with water and air (1); Biological: roots, animals and microbes (1).	1+1+1
	OR: Barchan: crescent, limited sand, one wind direction (1); Longitudinal: long ridges parallel to wind (1); Different wind patterns (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Waves undercut cliffs (1); Wave-cut platforms (1); Caves (1); Arches and stacks (1); Beaches protect the coast (1).	1+1+1+1+1
	OR: Glacial lake held by ice or moraine (1); Melting or heavy rain strains dam (1); Sudden release floods valleys (1); Chamoli flood in February 2021 (1); Monitoring lakes reduces risk (1).	1+1+1+1+1
20	U-shaped valleys and cirques (1); Arete and hanging valleys (1); Fjords (1); Moraines (1); Tourism, harbours and fresh water (1).	1+1+1+1+1
	OR: Acidic water dissolves limestone (1); Caves and underground rivers (1); Stalactites and stalagmites (1); Sinkholes (1); Water, tourism and cultural value (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).
- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.
- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).

- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.
- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND